

Chapter 5: The Real Numbers

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A: Cauchy Sequences

Definition 1 - (Eventual) ε -steadiness

Let $\varepsilon > 0$. A sequence $(a_n)_{n=0}^\infty$ is ε -steady if and only if for all natural numbers j, k , $d(a_j, a_k) \leq \varepsilon$. A sequence $(a_n)_{n=0}^\infty$ is eventually ε -steady if and only if the subsequence $(a_n)_{n=N}^\infty$ is ε -steady for some $N \in \mathbb{N}$.

Problem 1 - Examples and Non-examples of (eventual) ε -steadiness

We show examples and non-examples of (eventual) ε -steady sequences.

Proposition 1.1. *The sequence $(a_n) = 10^{-n} = 0.1, 0.01, 0.0001, \dots$ is 0.1-steady but not 0.01-steady.*

Proof. For the sequence $(a_n) = 10^{-n} = 0.1, 0.01, 0.0001, \dots$, observe that for any positive natural numbers j, k we have that $0 \leq 10^{-j}, 10^{-k} \leq 0.1$. Notice that $|10^{-j} - 10^{-k}| \leq \max\{10^{-j}, 10^{-k}\} \leq 0.1$. Hence, the sequence is 0.1-steady.

However, the sequence is not 0.01-steady because $|a_1 - a_2| = |0.1 - 0.01| = 0.09 \geq 0.01$. $\square$

Proposition 1.2. *The sequence $(a_n) = 2^n = 1, 2, 4, 8, \dots$ is not ε -steady for any $\varepsilon > 0$.*

Proof. Suppose for the sake of a contradiction that the sequence $(a_n) = 2^n = 1, 2, 4, 8, \dots$ is ε -steady for some $\varepsilon_0 > 0$. Then, it would imply that for all positive numbers j, k , we have $|2^j - 2^k| \leq \varepsilon_0$. Pick $k = 0$, so $|2^j - 1| \leq \varepsilon_0$ implies $2^j \leq 1 + \varepsilon_0$. Then, by Proposition 4.4.1, we can find a natural number N such that $1 + \varepsilon_0 < N$. Hence, we have that $2^j \leq 1 + \varepsilon_0 < N$, so $2^j < N$. If we choose $j := N$, we have that $2^N < N$, which contradicts the fact that $2^N \geq N$ for all natural numbers N . $\square$

Proposition 1.3. *The sequence $10, 0, 0, 0, \dots$ is not ε -steady for any $\varepsilon < 10$, but is eventually ε -steady for $\varepsilon > 0$.*

Proof. The sequence $10, 0, 0, 0, \dots$ is not ε -steady for any $\varepsilon < 10$ because we have that $|10 - 0| = 10 \not\leq \varepsilon$. However, the sequence is eventually ε -steady if we choose $N = 1$. This is because for all $j, k \geq 1$, we have $|a_j - a_k| = 0 < \varepsilon$. $\square$

Definition 2 - Cauchy Sequences

A sequence $(a_n)_{n=0}^\infty$ is a Cauchy sequence if and only if for all $\varepsilon > 0$, the sequence is eventually ε -steady. In other words, $\forall \varepsilon > 0, \exists N \in \mathbb{N}$ such that $j, k \in \mathbb{N} \implies d(a_j, a_k) \leq \varepsilon$.

Definition 3 - Boundedness

Let $M \geq 0$ be rational. A finite sequence $a_1, a_2, \dots, a_n$ is bounded by M if and only if $|a_i| \leq M$ for all $1 \leq i \leq n$. An infinite sequence is bounded by M if and only if $|a_i| \leq M$ for all $i \geq 0$. Overall, a sequence is said to be bounded if and only if it is bounded by M for some rational $M \geq 0$.

Proposition 1.4. *The sequence $(a_n)_{n=0}^\infty = (-1)^n \times n = 1, -2, 3, -4, \dots$ is unbounded.*

Proof. Suppose the sequence $(a_n)_{n=0}^\infty = (-1)^n \times n = 1, -2, 3, -4, \dots$ is bounded, so there exists a rational number $M \geq 0$ such that $|a_n| = n \leq M$ for all $n \geq 1$. By Proposition 4.4.1, there exists a natural number N such that $N > M$. But by our supposition, we also have that $|a_N| = N \leq M$, a contradiction! $\square$

Problem 2 - Cauchy sequences are bounded

Every Cauchy sequence $(a_n)_{n=0}^\infty$ is bounded.

Proof. Suppose the sequence $(a_n)_{n=0}^\infty$ is a Cauchy sequence, so we choose $\varepsilon = 1$. Then, there exists a natural number N such that for all $j, k \geq N$, we have $d(a_j, a_k) = |a_j - a_k| \leq 1$. Then, by the reverse triangle inequality, $||a_j| - |a_k|| \leq |a_j - a_k| \leq 1$. By simplifying, we get that $|a_j| \leq 1 + |a_k|$. Pick $k := N$ so that for any $j \geq N$, we have $|a_j| \leq 1 + |a_N|$. Therefore, for the subsequence of $(a_n)_{n=N}^\infty$, it is bounded by $1 + |a_N|$. Then, for the finite subsequence $(a_n)_{n=0}^N$, there exists M such that M bounds the finite subsequence. Therefore, the whole sequence is bounded by $\max\{1 + |a_N|, M\}$. $\square$

B: Equivalent Cauchy Sequences

Definition 4 - Equivalent Sequences

Two sequences $(a_n)_{n=0}^\infty$ and $(b_n)_{n=0}^\infty$ are equivalent if and only if for every $\varepsilon > 0$, the sequences are eventually ε -close. In other words, $\exists N \in \mathbb{N}$ such that $\forall n \geq N$, $|a_n - b_n| < \varepsilon$.

Problem 1 - Equivalence of Cauchy Sequences

If $(a_n)_{n=0}^\infty$ and $(b_n)_{n=0}^\infty$ are equivalent sequences of rationals, then $(a_n)_{n=0}^\infty$ is a Cauchy sequence if and only if $(b_n)_{n=0}^\infty$ is a Cauchy sequence.

Proof. Suppose $(a_n)_{n=0}^\infty$ and $(b_n)_{n=0}^\infty$ are equivalent sequences of rationals, and suppose also $(a_n)_{n=0}^\infty$ is a Cauchy Sequence. We now show that $(b_n)_{n=0}^\infty$ is also Cauchy. Let $\varepsilon > 0$ be given.

By the equivalence of the two sequences, there exists a natural number N such that if $n \geq N_1$, then $d(b_n, a_n) \leq \frac{\varepsilon}{4}$. Moreover, we also have by $(a_n)_{n=0}^\infty$ being Cauchy that there exists a natural number M such that if $j, k \geq M$, then $d(a_j, a_k) \leq \frac{\varepsilon}{2}$. We choose $L = \max\{N, M\}$ so that we have the useful properties from equivalence and Cauchy sequences. Then, by one of the properties of the distance function (specifically of “If $d(x, y) \leq \varepsilon$ and $d(z, w) \leq \delta$, then $d(x \pm z, y \pm w) \leq \varepsilon + \delta$ ”), we have that

$$\begin{aligned}
d(b_j, a_j) \leq \frac{\varepsilon}{4} \text{ and } d(b_k, a_k) \leq \frac{\varepsilon}{4} &\implies d(b_j - b_k, a_j - a_k) \leq \frac{\varepsilon}{2} && \text{by properties of the distance function} \\
&\implies |(b_j - b_k) - (a_j - a_k)| \leq \frac{\varepsilon}{2} && \text{by definition of distance} \\
&\implies ||b_j - b_k| - |a_j - a_k|| \leq \frac{\varepsilon}{2} && \text{by the reverse triangle inequality} \\
&\implies |b_j - b_k| - |a_j - a_k| \leq \frac{\varepsilon}{2} && \text{because we only care about being } \leq \frac{\varepsilon}{2} \\
&\implies |b_j - b_k| - \frac{\varepsilon}{2} \leq \frac{\varepsilon}{2} && \text{by } (a_n)_{n=0}^\infty \text{ being a Cauchy sequence} \\
&\implies d(b_j, b_k) \leq \varepsilon && \text{by rearrangement.}
\end{aligned}$$

For the converse, just switch the two sequences. $\square$

Problem 2 - Boundedness of Equivalent Cauchy Sequences

Let $\varepsilon > 0$. Show that if $(a_n)_{n=0}^\infty$ and $(b_n)_{n=0}^\infty$ are eventually ε -close, then $(a_n)_{n=0}^\infty$ is bounded if and only if $(b_n)_{n=0}^\infty$ is bounded.

Proof. Let $\varepsilon > 0$ be given. Suppose the sequences $(a_n)_{n=0}^\infty$ and $(b_n)_{n=0}^\infty$ are eventually ε -close, so they are equivalent sequences. Suppose also that $(a_n)_{n=0}^\infty$ is bounded, so there exists a non-negative rational M such that $|a_i| \leq M$ for all $i \geq 0$. We wish to show that $(b_n)_{n=0}^\infty$ is also bounded.

Because the two sequences are eventually ε -close, without loss of generality, we arbitrarily picked $\varepsilon := 1$. We then choose N such that for all $n \geq N$, we have that $|b_n - a_n| \leq 1$. Then, by the reverse triangle inequality, we have that $||b_n| - |a_n|| \leq |b_n - a_n| \leq 1$. By transitivity, we get $||b_n| - |a_n|| \leq 1$. By the properties of the absolute value, we have that $-1 \leq |b_n| - |a_n| \leq 1$. Focusing on the middle and right hand side of the inequalities, we get $|b_n| - |a_n| \leq 1$. Adding $|a_n|$ to both sides gives $|b_n| \leq 1 + |a_n|$. Because of $(a_n)_{n=0}^\infty$ being bounded, we have that $|a_n| \leq M$. By substitution, we get $|b_n| \leq 1 + M$. Therefore, for the tail of the infinite subsequence $(b_n)_{n=N}^\infty$, it is bounded by $M + 1$. For the head of the finite subsequence, it is bounded by some $L := \max\{|b_0|, |b_1|, \dots, |b_{N-1}|\}$. Hence, the overall subsequence $(b_n)_{n=0}^\infty$ is bounded by $\max\{1 + M, L\}$.

For the converse, just switch the sequences being treated. The logic follows because the distance function is symmetric. $\square$